

Risk-Adaptive Evidence Revalidation for Consequential Tool Actions: A Prospective Failure Analysis

Shaik Khaja Nayab Rasool

Moinabad Road, near the TS Police Academy, Aziznagar, Hyderabad, Telangana 500075, India
shaik.khajanayabrasool@gmail.com

Abstract. Consequential tool actions often depend on evidence that was valid when collected but may be invalid when used. Revalidating every prerequisite is costly, while acting on stale authorization, policy, identity, or operational state can cause harm. We formulate budgeted evidence revalidation as joint selection and abstention rather than as an evaluation of software tools. A preregistered first method (RAER v1) used a hard residual-risk feasibility threshold and failed its safe-completion gate. We then prospectively froze RAER v2, which minimizes an interpretable expected-loss objective combining harmful action, unnecessary abstention, validation cost, bounded budget slack, and an authorization safeguard. Six-domain leave-one-domain-out selection was performed on 72 exposed design cases; a separate 24-case test partition remained sealed. Out of fold, RAER v2 completed 25 of 27 valid cases (92.6%), caused harmful action in 14 of 45 invalid cases (31.1%), and incurred mean validation cost 0.547. FIXED 0.20 completed 27 of 27 valid cases, caused 18 harmful actions (40.0%), and cost 0.800. Seven of eight prospective criteria passed, but the required safe-completion rate was at least 95%. The gate therefore failed and no held-out evaluation was run. The result shows why favorable harm-cost summaries are insufficient when configuration transfer across domains degrades completion, and demonstrates a reproducible protocol for preserving negative design evidence without consuming confirmatory data.

Keywords: evidence revalidation · abstention · authorization · tool actions · validation budget · prospective evaluation

1 Introduction

Tool-using agents do more than produce text: they may change access, release records, modify financial instructions, update employee status, or commit an operational transaction. Benchmarks for tool agents consequently expose state dependencies, policy constraints, and potentially severe failures [11-14]. A recurring but less isolated problem is temporal: an action can be planned from evidence that was once correct yet is no longer fit for use. An authorization may have been revoked, a governing policy may have changed, an identity binding may have expired, or a resource may have moved after the evidence was observed.

The obvious control—revalidate every prerequisite—can be infeasible when authoritative checks consume latency, money, rate limits, or human attention. The alternative—reuse all stored evidence—can allow invalid prerequisites to survive until an irreversible action is executed. This creates a decision problem with three linked choices: which evidence to check, whether to act after valid checks, and when to request authority or abstain. The central quantity is not generic model confidence.

It is the expected operational loss induced by the remaining invalidity of action-specific prerequisites under a validation budget.

Risk-Adaptive Evidence Revalidation (RAER) was developed to study this problem with deterministic scenario states and explicit validation costs. RAER v1 selected a minimum-cost subset satisfying a hard residual-risk threshold. Its registered validation run reduced harmful actions and cost relative to a fixed 0.20 invalidity threshold, but safe completion was only 8/11 (72.7%) versus 11/11 for the best comparator. The protocol therefore required a stop before the held-out partition. Rather than relaxing the gate after observing the result, we retained v1 as a failure analysis and prospectively specified RAER v2.

RAER v2 replaces the hard feasibility rule with a joint expected-loss objective, introduces a mandatory safeguard for sufficiently risky authorization evidence, permits only a small penalized budget-boundary allowance, and enforces completion at an aggregate selection gate. This manuscript reports the complete design-stage result. It is intentionally not a positive effectiveness paper: v2 passed seven of eight criteria but completed only 25/27 valid cases (92.6%), below the prespecified 95% requirement. Consequently, the untouched 24-case test labels were not released.

The contribution is deliberately narrow. Budgeted evidence acquisition, abstention, stale-memory evaluation, and tool-safety enforcement already exist as separate and sometimes combined research themes. In particular, Budgeted Conformal Evidence Acquisition (BCEA) already couples additional evidence acquisition with abstention under a budget for vision-language claims [4]. We therefore do not claim that budgeted acquisition or abstention is new. The contribution evaluated here is their action-prerequisite instantiation: evidence validity changes through deterministic state transitions; evidence checks have explicit costs; action consequence and irreversibility affect modeled harm; authorization evidence receives a prespecified safeguard; and a prospective gate can preserve an informative negative result.

The paper makes four contributions:

- A formal, interpretable objective for selecting action-prerequisite checks while choosing between action and abstention under a bounded validation budget.
- An authorization-sensitive safeguard and an auditable treatment of correlated prerequisites and small budget-boundary slack.
- A leakage-resistant, six-domain constructed benchmark protocol that separates reviewer-visible facts from investigator-only validity labels and preserves an untouched test partition.
- A prospectively registered negative design result identifying cross-domain completion instability despite favorable descriptive harm and cost results.

2 Related Work and Novelty Boundary

2.1 Selective Prediction and Costed Acquisition

Reject-option and selective-classification research formalizes when a predictor should abstain, commonly through a rejection cost or a risk-coverage constraint [1]. Active feature acquisition instead decides which observations to purchase before predicting. Joint approaches learn acquisition and prediction together to balance accuracy against

feature cost [2], while active-acquisition POMDPs extend this trade-off to sequential decisions under partial observability [3]. These lines establish that abstention and costed information acquisition are not individually novel.

BCEA is the closest located work at the component-combination level [4]. It gives a vision-language system three choices—answer, acquire further visual evidence, or abstain—and incorporates acquisition into a conformal calibration procedure. RAER differs in target and semantics rather than in the generic acquire-or-abstain pattern. BCEA re-examines visual claims to control hallucination risk; RAER revalidates named authorization, policy, scope, and state prerequisites before an external action. The current RAER benchmark uses deterministic state transitions and costed authoritative checks, not conformal coverage guarantees. This distinction is substantive, but it does not justify claiming the general combination of evidence acquisition and abstention as new.

2.2 Stale Memory and Agentic Abstention

Recent memory benchmarks directly examine whether agents recognize that stored information is no longer valid. STALE constructs implicit conflicts in which later observations invalidate earlier memories and tests state resolution, premise resistance, and policy adaptation [5]. MemConflict treats memory fitness for use as query-conditioned across temporal, factual, and contextual conflicts [6]. These works motivate explicit validity modeling but evaluate memory retrieval and answer behavior rather than the selection of authoritative checks before consequential state change.

AgentAbstain evaluates paired should-act and should-abstain tasks in executable sandboxes [7], while agentic abstention treats stopping as a sequential decision when further interaction is unlikely to help [8]. These benchmarks prevent RAER from claiming novelty for ACT/ABSTAIN pairing or timely stopping. RAER's narrower distinction is that REFRESH and ASK are operational consequences of selected prerequisite checks, with cost and authorization semantics attached to each check.

2.3 Runtime Verification and Tool-Agent Safety

Contract and policy systems gate tool calls against formal requirements. ToolGate maintains a symbolic state and uses preconditions and postconditions to control invocation and state commitment [9]. Solver-aided policy verification translates operational policies into constraints checked before tool execution [10]. These methods establish the importance of explicit prerequisites. RAER assumes, instead, that an observable prerequisite can itself be stale and that not all observations can be refreshed within budget; its selector decides which facts to reacquire before the contract is trusted.

ToolEmu supports scalable risk discovery through an LM-emulated sandbox [11]. The tau-bench and ToolSandbox benchmarks introduce dynamic conversation, domain policy, stateful tools, and implicit dependencies [12, 13]. Agent-SafetyBench broadens coverage across environments and safety failures [14]. ToolSafe and SafeAgent add proactive intervention and stateful runtime protection, particularly for adversarial threats [15, 16]. RAER does not compete as a generic prompt-injection

detector or tool guardrail. Its primary threat model is non-adversarial evidence invalidity between observation and action.

Table 1. Closest research families and the remaining RAER-specific boundary

Research family	Established capability	RAER-specific boundary
Selective prediction	Risk–coverage or reject-cost trade-off	Checks named action prerequisites before deciding
Active acquisition	Costed selection of additional features	Revalidates previously observed evidence after state drift
BCEA	Budgeted acquire/answer/abstain	Authorization, tool-action harm, deterministic mutable state
Stale memory	Detects obsolete or conflicting memories	Authoritative refresh chosen under an action budget
Agent abstention	Should-act/should-abstain or timely stopping	REFRESH/ASK arise from prerequisite verification
Contracts/policy	Precondition and policy enforcement	Treats prerequisite observations as uncertain and costly
Tool-safety benchmarks	Risk and stateful-tool evaluation	Isolates non-adversarial evidence invalidity and check cost

Novelty claim. Within the screened literature, we did not locate an identical formulation joining mutable action prerequisites, consequence-weighted residual harm, explicit validation budgets, mandatory authorization checks, and a prospectively enforced safe-completion gate. This is a bounded literature-audit conclusion, not a claim of global or patent-level novelty.

3 Problem Formulation

A scenario contains a proposed action, a validation budget B , an action consequence score C , an irreversibility score I , and m evidence prerequisites. The present benchmark uses $m=3$. For evidence i , q_i is a pre-action invalidity estimate, $w_i=W_i/5$ is normalized criticality, and $c_i=K_i/5$ is normalized validation cost. C , I , W_i , and K_i are rubric scores from 1 to 5. Actual validity is an investigator-only outcome and cannot be used to select checks.

The frozen $E0$ estimator converts only reviewer-visible facts into q_i . Its inputs are normalized evidence age, source volatility, dependent changes, update frequency, inverse source reliability, authorization age, and contradictory observations. With feature vector x_i and fixed coefficient vector $\beta=(1.25, 0.85, 0.75, 0.45, 0.95, 1.15, 1.40)$, $E0$ is:

$$q_i = \text{sigmoid}(-3.40 + \beta^T x_i). \quad (1)$$

$E0$ was selected during calibration using mean leave-one-domain-out Brier score and the one-standard-error rule. It was not refitted for $v2$. Action harm combines consequence with irreversibility:

$$h = (C/5)[0.5 + 0.5(I-1)/4]. \quad (2)$$

For a candidate validation subset S , $q'_i=0.01$ when i is checked and returns valid; otherwise $q'_i=q_i$. The 0.01 floor represents residual verifier error. Modeled residual harm is:

$$R(S) = \min\{h, h[1 - \prod_i (1 - q'_i w_i)] + U(S)\}. \quad (3)$$

$U(S)$ is a frozen correlation uplift. If at least two unchecked prerequisites belong to the predeclared correlated group, $U(S)=0.20h$ times the maximum unchecked-pair value $\sqrt{q_i q_j} \min(w_i, w_j)$; otherwise $U(S)=0$. This dependence term is applied once. The safe-probability proxy is:

$$P_{\text{safe}}(S) = \prod_i (1 - q'_i). \quad (4)$$

$R(S)$ and $P_{\text{safe}}(S)$ are planning proxies, not observed outcomes or guarantees. With $\lambda_H=1$ and slack penalty $\lambda_S=1$, the v2 objective is:

$$J(S) = \lambda_C \sum_{i \in S} c_i + \lambda_S \max(0, \sum_{i \in S} c_i - B) + \min\{\lambda_{HR}(S), \lambda_{AP_safe}(S)\}. \quad (5)$$

Candidate subsets satisfy $\sum c_i \leq B + \delta$. Exact enumeration chooses the minimum $J(S)$, breaking ties by lower validation cost, lower residual harm, fewer checks, and lexicographic evidence identifiers. After every selected check returns valid, the policy acts when $\lambda_{HR}(S) \leq \lambda_{AP_safe}(S)$; otherwise it abstains. A detected invalid authorization returns ASK, whereas another detected invalid prerequisite returns REFRESH. Either detected invalidity overrides the planned action decision.

3.1 Authorization Safeguard

When authorization sensitivity is at least 4 and an authorization prerequisite satisfies $q_{iw_i} \geq \theta_{\text{auth}}$, that prerequisite is mandatory. If all mandatory authorization checks do not fit within $B + \delta$, the policy abstains. It may not omit authorization silently. This rule reflects a qualitative distinction: many stale-state failures can be repaired by refresh, while missing authority can make the action impermissible regardless of other evidence.

3.2 Controlled Boundary Slack

The candidate δ values are 0 and 0.05. Because each full rubric cost unit equals 0.20 after normalization, the 0.05 allowance cannot purchase a complete additional unit. Slack is penalized one-for-one in $J(S)$, and the protocol caps its case rate, mean, and maximum. It is therefore a controlled boundary device rather than an unrestricted budget relaxation.

4 Benchmark and Integrity Controls

4.1 Scenario Structure

The frozen RAER-B96 benchmark contains 96 constructed cases across commerce, cybersecurity, finance, healthcare administration, human resources, and privacy, with 16 cases per domain. Each case contains three evidence prerequisites and one proposed consequential action. Cases span all-valid states, single invalidities, correlated failures, independent dual failures, authorization revocation, policy change,

contradiction, and budget insufficiency. The original split contains 48 development, 24 validation, and 24 held-out cases.

Each reviewer-visible record contains the operational context, proposed action, evidence descriptions, evidence kind, age, validity window, volatility, update frequency, source reliability, dependent changes, contradiction strength, validation check, and scenario budget. Investigator-only files contain actual validity, correlation membership, and state-transition labels. This physical separation prevents a reviewer or selector from inferring q_i from the observed outcome by construction.

4.2 Scoring and Review

Consequence, irreversibility, authorization sensitivity, evidence criticality, and validation cost were scored using frozen 1–5 rubrics with boundary examples. Two isolated model-based pre-reviewers independently scored the reviewer-visible package with different safety-governance and operations-reliability orientations. Disagreements were preserved and adjudicated in a versioned register. This process was used to stress-test clarity and rubric consistency; it is not human inter-rater reliability and is not presented as a substitute for independent domain experts.

Table 2. Benchmark scoring dimensions

Dimension	Unit	Operational meaning
Consequence C	1–5	Severity if the proposed action is harmful
Irreversibility I	1–5	Difficulty of undoing or containing the action
Authorization sensitivity	1–5	Dependence on current, explicit authority
Criticality W_i	1–5	How strongly invalid evidence undermines safe action
Validation cost K_i	1–5	Relative authoritative-check burden

4.3 Leakage, Provenance, and Version Controls

Candidate benchmark versions underwent blinded clarity review, identifier-pattern checks, duplicate-text checks, topology-order randomization, forbidden-clue scans, schema validation, and exact identifier coverage tests. The final reviewer-visible release and adjudicated scores were locked with SHA-256 hashes. Pilot artifacts, scoring outputs, source provenance, experiment versions, and decision gates remain immutable. Later closure manifests are additive and do not rewrite the original release lock.

The held-out labels are represented by an absent release file. Both v1 and v2 evaluators abort if that file is present before authorization. At the time of this manuscript, the file remains absent. This negative control is important: a held-out set is not protected merely because results are omitted; it must be inaccessible to design and selection code.

5 Method History and Prospective Design

5.1 RAER v1 as the Motivating Failure

RAER v1 used a consequence-adjusted hard threshold. It selected the cheapest subset whose residual risk was below $\tau_0(1-\alpha h)$, with $\tau_0=0.22$ and $\alpha=0.30$; when no affordable subset was feasible, it selected the minimum-risk affordable subset and abstained. On the 24-case registered validation partition, v1 caused 3/13 harmful actions (23.1%) versus 4/13 (30.8%) for FIXED_0.20 and reduced mean cost from 0.850 to 0.333. However, it completed only 8/11 valid cases (72.7%) versus 11/11 for the comparator. The preregistered decision was STOP_BEFORE_HELD_OUT. These values motivated v2 but are not relabeled as v2 evidence.

5.2 Prospective RAER v2 Grid

Before any aggregate v2 result, the objective, grid, implementation, tests, design plan, and 72 exposed label inputs were hashed into a nine-file pre-execution lock. The harmful-action and slack weights remained fixed at one. The Cartesian grid contained 80 configurations.

Table 3. Prospectively frozen RAER v2 configuration grid

Parameter	Values	Role
λ_H	1.0	Harmful-action loss
λ_A	0.25, 0.50, 0.75, 1.00, 1.50	Unnecessary-abstention loss
λ_C	0.02, 0.05, 0.10, 0.20	Validation-cost loss
δ	0, 0.05	Maximum boundary slack
θ_{auth}	0.08, 0.12	Authorization trigger
λ_S	1.0	Slack penalty

5.3 Nested Design-Only Evaluation

Because the v1 development and validation labels had been exposed, all 72 cases were treated as non-confirmatory design data. Six outer folds each omitted one complete domain. Within each fold, all 80 configurations were evaluated on the other five domains. A configuration was eligible only if it met the safe-completion, harmful-action, authorization, and slack limits. Eligible configurations were ordered by lower harm, lower mean cost, higher completion, lower slack rate, and frozen parameter tie-breaks. The chosen configuration was then applied unchanged to the omitted domain. Concatenating outer predictions yielded one out-of-fold result for every design case.

The registered comparators were STATIC, CONTRACT_ONLY, FIXED_0.10, FIXED_0.20, FIXED_0.35, FIXED_0.50, ALWAYS_REFRESH, and frozen RAER v1. FIXED_t checks evidence with $q_i \geq t$. CONTRACT_ONLY checks authorization and scope evidence. ALWAYS_REFRESH checks all evidence. Comparator checks were retained even when their rule exceeded a scenario budget, with cost and overrun reported; RAER itself remained subject to $B+\delta$.

5.4 Gate and Uncertainty

Table 4. Prospective v2 gate; every criterion was mandatory

Criterion	Required value
Safe completion	Within 0.05 of the best comparator
Harmful action	No worse than FIXED_0.20
Authorization	Zero triggered-authorization harmful actions
Non-dominance	No comparable-safe policy no worse in harm and cost, one strict
Positive-slack rate	≤ 0.25
Mean slack	≤ 0.025
Maximum slack	≤ 0.05
Fold stability	Eligible configuration in at least five of six folds

Primary outcomes were safe successes divided by all-valid cases, harmful actions divided by cases with at least one invalid prerequisite, and mean validation cost across all cases. Numerators and denominators were fixed. Ten thousand scenario bootstraps stratified by domain used seed 20260812 and percentile 95% intervals. These intervals are descriptive because all 72 cases are design data; they cannot turn the result into confirmatory evidence.

6 Results

6.1 Main Out-of-Fold Result

RAER v2 safely completed 25 of 27 all-valid cases and blocked 31 of 45 invalid cases. Fourteen invalid cases proceeded to harmful action, and two valid cases were false blocks. Mean validation cost was 0.547. FIXED_0.20 completed all 27 valid cases, blocked 27 invalid cases, caused 18 harmful actions, and cost 0.800. Thus v2 showed a favorable descriptive movement in harm and cost but lost two valid completions.

Table 5. Main design-stage results on 72 exposed cases

Policy	Safe completion	Harmful action	Mean cost	False blocks
STATIC	27/27 (1.000)	45/45 (1.000)	0.000	0
CONTRACT_ONLY	27/27 (1.000)	10/45 (0.222)	0.956	0
FIXED_0.10	27/27 (1.000)	0/45 (0.000)	1.319	0
FIXED_0.20	27/27 (1.000)	18/45 (0.400)	0.800	0
FIXED_0.35	27/27 (1.000)	40/45 (0.889)	0.150	0
FIXED_0.50	27/27 (1.000)	45/45 (1.000)	0.000	0
ALWAYS_REFRESH	27/27 (1.000)	0/45 (0.000)	1.456	0
RAER v1	16/27 (0.593)	16/45 (0.356)	0.364	11
RAER v2 OOF	25/27 (0.926)	14/45 (0.311)	0.547	2

For RAER v2, descriptive domain-stratified bootstrap intervals were 0.815–1.000 for safe completion, 0.182–0.449 for harmful-action rate, and 0.481–0.611 for mean validation cost. FIXED_0.20 intervals were 1.000–1.000, 0.257–0.545, and 0.681–0.922, respectively. The overlap in harmful-action uncertainty and the small valid-case denominator preclude a claim of superiority.

6.2 Prospective Gate Decision

Seven criteria passed. RAER v2 had no comparable-safe dominator, harmful-action rate was no worse than FIXED_0.20, no harmful action arose from an unchecked triggered authorization prerequisite, positive slack occurred in 12/72 cases (16.7%), mean slack was 0.0083, maximum slack was 0.05 up to floating-point tolerance, and all six outer folds selected an eligible configuration. Safe completion was the only failure: the best comparator achieved 1.00, so the gate required at least 0.95; v2 achieved 25/27=0.9259.

Table 6. Registered gate outcome

Criterion	Observed	Decision
Safe completion	0.9259; required ≥ 0.9500	FAIL
Harmful action	0.3111 vs. FIXED_0.20 at 0.4000	PASS
Authorization	0 triggered-authorization harms	PASS
Non-dominance	No comparable-safe dominator	PASS
Positive-slack rate	0.1667; limit 0.25	PASS
Mean slack	0.0083; limit 0.025	PASS
Maximum slack	0.0500; limit 0.05	PASS
Fold stability	6/6 eligible folds	PASS

The registered decision was FAIL_KEEP_HELD_OUT_SEALED. With 27 valid cases, one fewer false block would have produced 26/27=96.3%, but discreteness does not permit changing the threshold or substituting the fitted-design result. The test release file was not created, and no held-out outcome is reported.

6.3 Failure Analysis

The two out-of-fold false blocks were RB-FIN-04 and RB-PRI-05. In both the finance and privacy outer folds, the inner selector chose $\lambda_A=0.50$ rather than 1.00. For RB-FIN-04, one check costing 0.60 left modeled residual harm 0.391 and safe-probability proxy 0.447; the policy abstained. For RB-PRI-05, the selected subset was empty, with residual harm 0.311 and safe-probability proxy 0.603; the policy again abstained. Every prerequisite in both cases was actually valid, but that information was unavailable to the selector.

Table 7. Out-of-fold false blocks

Case	Domain	λ_A	Checks / cost	Planning proxies	Outcome
RB-FIN-04	Finance	0.50	1 / 0.60	R=0.391; P_safe=0.447	ABSTAIN
RB-PRI-05	Privacy	0.50	0 / 0.00	R=0.311; P_safe=0.603	ABSTAIN

After the gate, the identical selector fitted to all 72 design cases chose $\lambda_A=1.00$, $\lambda_C=0.10$, $\delta=0.05$, and $\theta_{auth}=0.08$. That fitted configuration completed 26/27 valid cases. This observation localizes the issue to cross-domain configuration transfer, but it is not a replacement endpoint: it uses every exposed label and was computed after the out-of-fold gate.

6.4 Ablations

Ablations were evaluated descriptively using the all-design fitted configuration. Removing abstention loss caused the policy to abstain on every valid case. Removing the authorization safeguard preserved completion but increased harmful actions to 19/45 and produced seven triggered-authorization harmful actions. Removing slack reduced safe completion to 24/27 while slightly lowering harm and cost. Removing the correlation uplift did not change aggregate outcomes in these cases, suggesting that its present implementation was behaviorally inactive at the fitted operating point.

Table 8. Design-data ablations of the fitted v2 configuration

Variant	Safe completion	Harmful action	Mean cost	Authorization on harms
Full v2	26/27 (0.963)	14/45 (0.311)	0.606	0
No abstention loss	0/27 (0.000)	0/45 (0.000)	0.358	0
No authorization safeguard	26/27 (0.963)	19/45 (0.422)	0.561	7
No slack	24/27 (0.889)	13/45 (0.289)	0.528	0
No correlation uplift	26/27 (0.963)	14/45 (0.311)	0.606	0

7 Discussion

7.1 What the Result Supports

The design result supports a limited statement: on the exposed cases, RAER v2 occupied a useful descriptive harm-cost position and eliminated the authorization failure category targeted by its safeguard, but it did not meet the registered completion requirement under domain-held-out selection. This is evidence about method behavior and evaluation discipline, not evidence that v2 is generally effective or deployment-ready.

The most informative finding is the separation between pooled fit and cross-domain transfer. All six inner selections were eligible, yet two outer domains each contributed a false block. The eligibility threshold was evaluated on pooled inner-domain valid

cases, allowing $\lambda_A=0.50$ to sit barely above the discrete completion boundary in training while failing on one valid case in the omitted domain. The safe-completion constraint therefore controlled average inner performance but not worst-domain stability.

Authorization behaved differently. The prespecified mandatory-check rule eliminated triggered-authorization harms out of fold, and its removal produced seven such harms in the fitted ablation. Although this comparison is non-confirmatory, the mechanism is directly interpretable: the unconstrained objective can prefer a cheaper subset that ignores authorization, whereas the safeguard prohibits that omission when sensitivity and q_{iw_i} cross the threshold.

7.2 Why the Negative Gate Matters

A conventional results narrative might emphasize the reduction from 18 to 14 harmful actions and the 31.6% lower mean validation cost relative to FIXED_0.20. That narrative would conceal the operational cost of two unnecessary blocks and would ignore that 92.6% fell below a threshold fixed before inspection. The all-required gate prevents a multi-objective method from being declared successful by selecting whichever outcome moved favorably after the fact.

Preserving the sealed test set is also a research asset. Opening the 24 labels after a failed design gate would yield an easily overinterpreted, low-powered result and would remove those cases from future confirmatory use. The negative design result instead documents a precise defect—cross-domain completion instability—without spending the remaining evaluation resource.

7.3 Implications for a Separately Registered Successor

The present data suggest three prospective changes, none of which is part of v2. A successor could enforce worst-domain or lower-confidence-bound completion during inner selection, separate the evidence-subset objective from the final act/abstain threshold, and penalize configuration instability across inner domains. Such changes must be registered as a new method and evaluated first on newly constructed design cases. Reusing the current 72 cases alone to tune until the gate passes would convert the reported failure into outcome-driven optimization.

The authorization safeguard should be retained unless new evidence contradicts its benefit. Bounded slack also appears consequential: removing it reduced completion by two additional valid cases, although it modestly improved harm and cost. The correlation uplift, by contrast, warrants either a more sensitive benchmark stratum or removal in a future registered ablation because its current aggregate effect was zero.

8 Threats to Validity

8.1 Construct Validity

Harm, criticality, cost, and authorization sensitivity are rubric-derived proxies. E0 is monotone and interpretable but is not a calibrated probability model for any deployed organization. The 0.01 verifier floor, 0.20 correlation uplift, and normalized K/5 costs are design conventions. Results therefore compare policies within the constructed

benchmark; they do not estimate real incident probability, financial loss, latency, or human workload.

8.2 Internal Validity

Reviewer-visible features and investigator labels were separated, selection was deterministic, and hashes bind code to inputs. Nevertheless, scenario authors determined the state transitions and rubric anchors. Pattern leakage was reduced through multiple candidate rewrites, identifier/topology checks, and blinded review, but cannot be ruled out completely. The two model-based reviewers may share training-data and reasoning biases; their agreement is AI–AI consistency, not human reliability.

8.3 External and Statistical Validity

The benchmark has six domains, 96 cases, and exactly three prerequisites per case. The v2 design result uses only 72 exposed cases, of which 27 are all-valid. With this denominator, completion changes in increments of 3.7 percentage points, and the descriptive bootstrap interval is wide. No operational deployment, external dataset, latency measurement, or independent human-domain evaluation is included. Generalization beyond the benchmark is unsupported.

8.4 Novelty and Publication Validity

The novelty audit screened 25 close works and was updated after locating BCEA, which narrows the contribution materially. The audit is not a systematic review, patent search, or guarantee of conceptual uniqueness. The manuscript reports v1 and v2 failures to avoid selective history. It must not be described as validating RAER, proving superiority, or establishing production readiness.

9 Reproducibility and Research Integrity

The research package preserves the scenario provenance register, reviewer-visible release, investigator vault, split manifest, scoring and adjudication outputs, v1 and v2 protocols, versioned evaluators, boundary tests, raw policy outcomes, bootstrap seed, pre-execution locks, and additive closure manifests. Eight v2 implementation tests cover configuration count, deterministic selection, slack equality and exceedance, authorization requirements, invalid-check override, action behavior, and summary denominators. The v2 design closure verifies 15 artifacts by SHA-256 and records that held-out labels were not accessed.

A public artifact should include reviewer-visible cases, schemas, rubrics, code, tests, released design labels where ethically and contractually permitted, raw design outcomes, and verification scripts. The coordinator workbook and unreleased test labels must remain restricted. File paths, usernames, organizational identifiers, and internal metadata must be removed before release. A claim-to-evidence ledger and citation log should accompany the camera-ready manuscript.

Acknowledgements

Use of generative AI. Generative AI systems assisted with scenario-clarity stress testing, code scaffolding, and manuscript organization. Model-based review outputs were treated as synthetic methodological checks, not human judgments or substitutes for domain-expert validation. The author reviewed the disclosed uses, remains responsible for verifying every scenario, result, citation, and statement, and accepts full accountability for the submitted version. No generative-AI image is included.

Disclosure of Interests

The author has no competing interests to declare that are relevant to the content of this article.

Funding

No funds, grants, or other external support were received.

Ethics Approval and Consent

This methodological study used constructed synthetic scenarios and did not involve human participants, identifiable personal data, animals, clinical interventions, or human-subject experimentation. Ethics approval, consent to participate, and consent for publication were therefore not applicable.

Author Contributions

Shaik Khaja Nayab Rasool: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization, and Project administration.

Code and Data Availability

The public artifact includes the reviewer-visible benchmark, exposed development and validation labels, schemas, reference code, tests, protocols, and design-stage results. It excludes the investigator label vault, restricted coordinator workbook, and sealed held-out labels. The artifact is licensed under the MIT License. Permanent repository: Zenodo; DOI/URL: To be assigned upon publication/deposit.

10 Conclusion

RAER v2 formalized pre-action evidence revalidation as an abstention-aware, authorization-sensitive selection problem under validation budgets. On exposed design data, it reduced harmful action and validation cost relative to FIXED_0.20 and eliminated the targeted authorization failure category. It nevertheless completed only 25 of 27 valid cases, below the prospectively required 95%. The method therefore failed its design gate, and the held-out partition remains sealed.

This outcome does not invalidate the research procedure. It refutes the prespecified claim that the current v2 selection procedure met all design requirements. The resulting contribution is a reproducible method and negative design-stage finding: favorable safety-cost summaries can coexist with insufficient cross-domain completion. A future version must address that instability prospectively on new design cases rather than revise v2 after observing its failure.

References

1. Franc, V., Prusa, D., Voracek, V.: Optimal strategies for reject option classifiers. *Journal of Machine Learning Research* 24(11), 1–49 (2023)
2. Shim, H., Hwang, S.J., Yang, E.: Joint active feature acquisition and classification with variable-size set encoding. In: *Advances in Neural Information Processing Systems* 31 (2018)
3. Li, Y., Oliva, J.: Towards cost sensitive decision making. In: *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics*. PMLR 258, 3601–3609 (2025)
4. Xu, J., Wu, Y., Zeng, D., Paisley, J., Zhao, Q.: Look again before you abstain: budgeted conformal evidence acquisition for reliable vision-language model. *arXiv:2606.16667* (2026). doi:10.48550/arXiv.2606.16667
5. Chao, H., Bai, Y., Sheng, R., Li, T., Sun, Y.: STALE: Can LLM agents know when their memories are no longer valid? *arXiv:2605.06527* (2026). doi:10.48550/arXiv.2605.06527
6. Tao, Z., et al.: MemConflict: Evaluating long-term memory systems under memory conflicts. *arXiv:2605.20926* (2026). doi:10.48550/arXiv.2605.20926
7. Liu, X., et al.: AgentAbstain: Do LLM agents know when not to act? *arXiv:2607.10059* (2026). doi:10.48550/arXiv.2607.10059
8. Luo, H., Wen, B., Wang, L.L.: Agentic abstention: Do agents know when to stop instead of act? *arXiv:2606.28733* (2026). doi:10.48550/arXiv.2606.28733
9. Liu, Y., et al.: ToolGate: Contract-grounded and verified tool execution for LLMs. *arXiv:2601.04688* (2026). doi:10.48550/arXiv.2601.04688
10. Winston, C., Winston, C., Just, R.: Solver-aided verification of policy compliance in tool-augmented LLM agents. *arXiv:2603.20449* (2026). doi:10.48550/arXiv.2603.20449
11. Ruan, Y., et al.: Identifying the risks of LM agents with an LM-emulated sandbox. *arXiv:2309.15817* (2024). doi:10.48550/arXiv.2309.15817
12. Yao, S., Shinn, N., Razavi, P., Narasimhan, K.: tau-bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv:2406.12045* (2024). doi:10.48550/arXiv.2406.12045
13. Lu, J., Holleis, T., Zhang, Y., et al.: ToolSandbox: A stateful, conversational, interactive evaluation benchmark for LLM tool use capabilities. *arXiv:2408.04682* (2025). doi:10.48550/arXiv.2408.04682
14. Zhang, Z., et al.: Agent-SafetyBench: Evaluating the safety of LLM agents. *arXiv:2412.14470* (2025). doi:10.48550/arXiv.2412.14470
15. Mou, Y., et al.: ToolSafe: Enhancing tool invocation safety of LLM-based agents via proactive step-level guardrail and feedback. *arXiv:2601.10156* (2026). doi:10.48550/arXiv.2601.10156
16. Liu, H., Ilyushin, E., Ni, J., Zhu, M.: SafeAgent: A runtime protection architecture for agentic systems. *arXiv:2604.17562* (2026). doi:10.48550/arXiv.2604.17562